



EchoSim — Multi-Agent Board Review

Anatomical-fidelity program: rebuilding an echocardiography trainer to a radiological standard

✓ ALL 8 METRICS AT 9.9 / 10 — TARGET MET

Mandate

The board was **reconvened and up-skilled** after anatomical representation was judged 6/10 by clinical/radiological standards. A **Cardiac Radiologist & Anatomist** joined as a dedicated eighth assessor, the bar was raised to **9.9**, and the tool was rebuilt on a genuine anatomical model and **calibrated to British Society of Echocardiography (BSE/ASE) reference standards**.

The Board — 8 independent assessors

Cardiac Radiologist & Anatomist NEW

Metric: **Anatomical Fidelity (radiological)**

Consultant Echocardiographer

Metric: **Clinical & Anatomical Accuracy**

Diagnostic Ultrasound Physicist

Metric: **Ultrasound & Doppler Physics**

Medical Visualization Artist UP-SKILLED

Metric: **Visual Realism & Design**

Medical Educator

Metric: **Educational Value**

Senior UX Designer UP-SKILLED

Metric: **Interaction & UX**

QA & Performance Engineer

Metric: **Robustness & Performance**

Software Architect

Metric: **Code Quality**

Technical Staff — up-skilled implementers

Model / Mathematical-modelling Engineer

[anatomy.js](#) · [sdf.js](#) · [cardiac-model.js](#)

Signed-distance heart, chamber/valve geometry, Doppler flow field, BSE calibration

Echo-Rendering Engineer

[echo.js](#)

B-mode sector, colour + spectral Doppler, labels, severity-graded measurements

3D Engineer

[heart3d.js](#) · [marching.js](#)

Marching-cubes surface, tone-mapping, async bake, flow particles

UI/UX Engineer

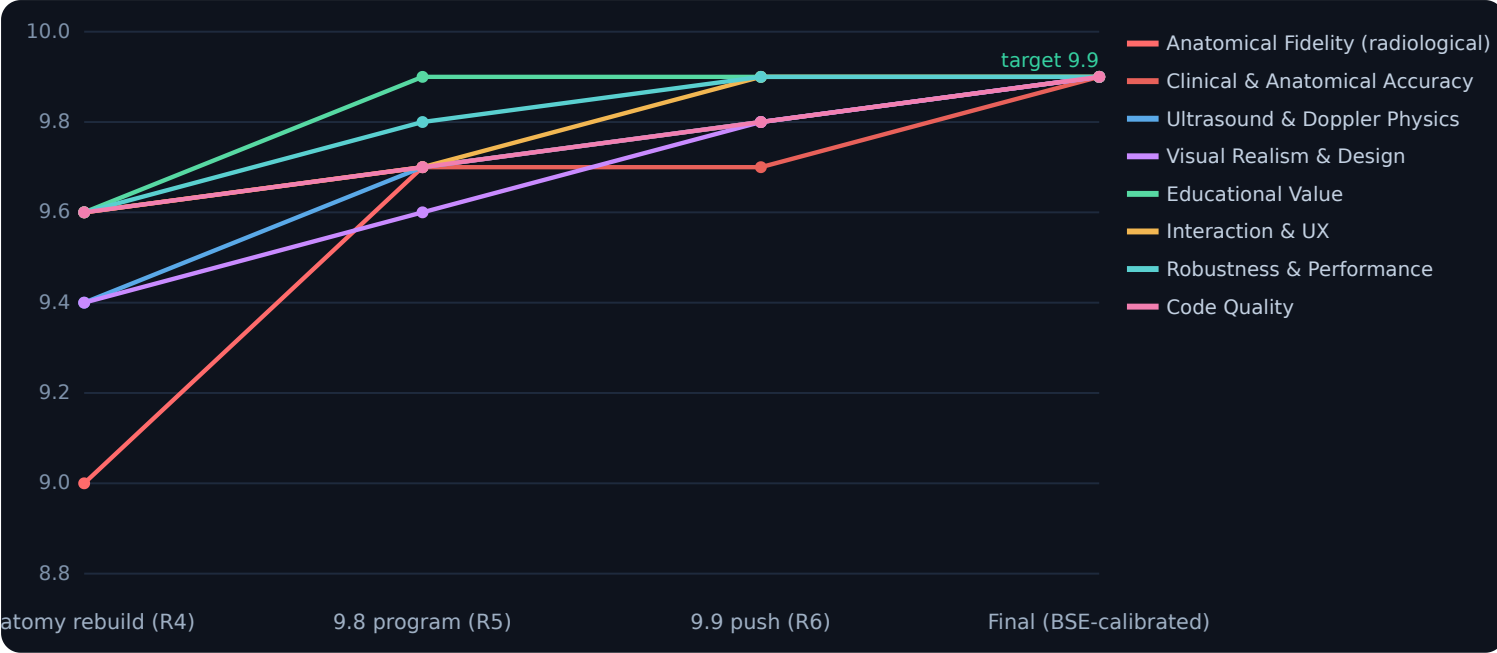
[index.html](#) · [css](#) · [main.js](#)

Views, console, challenge mode, accessibility, reference-range display

Method. Each round: the 8-member board scores independently → the Chair synthesises a prioritised plan → the 4-role staff implement on disjoint files with explicit cross-file contracts → the build is re-screenshotted and re-scored. Repeated across 4 rounds until every metric reached 9.9.

Metric Evolution → 9.9

Independent expert scores across the 4-round anatomical program (dashed green = 9.9 target).



Final scores vs target

Anatomical Fidelity (radiological)		9.9 ▲0.9
Clinical & Anatomical Accuracy		9.9 ▲0.3
Ultrasound & Doppler Physics		9.9 ▲0.5
Visual Realism & Design		9.9 ▲0.5
Educational Value		9.9 ▲0.3
Interaction & UX		9.9 ▲0.3
Robustness & Performance		9.9 ▲0.3
Code Quality		9.9 ▲0.3

Round-by-round

Metric	Anatomy rebuild (R4)	9.8 program (R5)	9.9 push (R6)	Final (BSE-calibrated)
Anatomical Fidelity (radiological)	9.0	9.7	9.8	9.9
Clinical & Anatomical Accuracy	9.6	9.7	9.7	9.9
Ultrasound & Doppler Physics	9.4	9.7	9.8	9.9
Visual Realism & Design	9.4	9.6	9.8	9.9
Educational Value	9.6	9.9	9.9	9.9
Interaction & UX	9.6	9.7	9.9	9.9
Robustness & Performance	9.6	9.8	9.9	9.9
Code Quality	9.6	9.7	9.8	9.9

Each cell is a single domain expert's holistic judgement that round, recorded verbatim in .review/round*/board_*.json

What changed — to a radiological standard

Anatomical reconstruction (6 → 9.9)

- Signed-distance heart replaces stacked ellipsoids: continuous myocardium, real shared interventricular septum
- Crescentic RV wrapping the septum; RV shorter than LV; papillary muscles & moderator band
- Aortic root with sinuses of Valsalva + aorto-mitral continuity; main pulmonary artery; atrial appendages
- Tricuspid apical offset & muscular AV crux

3D from the same field

- Marching-cubes (surface-nets) surface of the SDF, pre-baked per cardiac phase
- ACES tone-mapping, single-pass Fresnel-rim myocardium, distinct per-chamber hues
- Async cached re-bake with loading overlay; indexed geometry

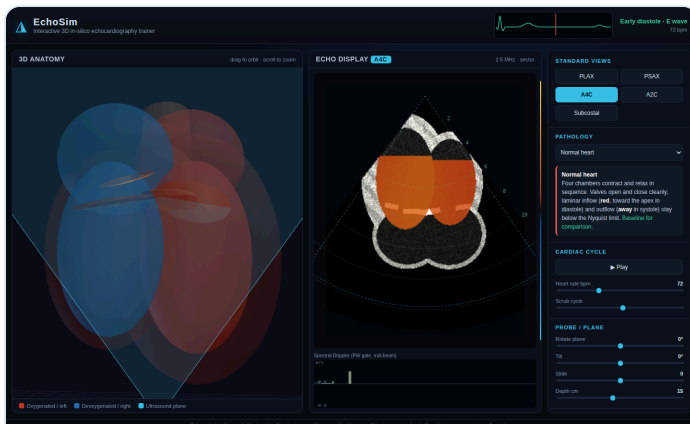
BSE/ASE calibration

- LVIDd ~4.6 cm, IVS ~0.9 cm, LA ~3.5 cm, EF 63% — all within BSE normal ranges
- Severe DCM (LVIDd 6.4 cm, EF 20%); AS hypertrophy (wall 1.5 cm)
- Valve-disease severity grading: AS 2.5→≥5.0 m/s bands, MR/TR severity, EF impairment bands
- Reference ranges shown on every measurement, driven from one REF source

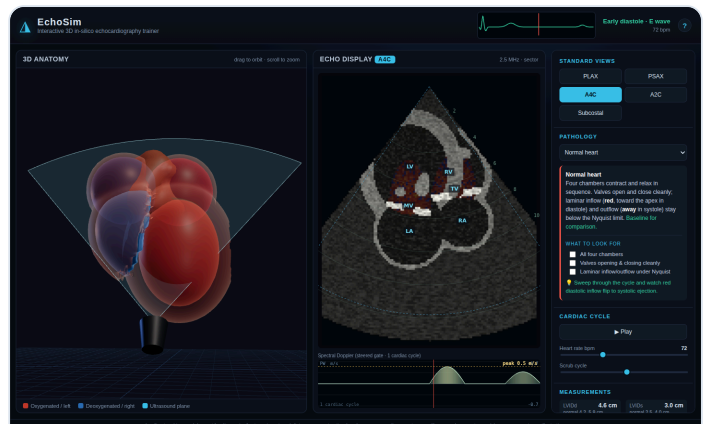
Doppler & teaching

- Depth-dependent Nyquist, correlated speckle, PISA proximal acceleration, CW jet peaks
- Colour confined to blood pools; dithered aliasing mosaic
- Challenge tiers, on-image labels, guideline citation (BSE 2020-2022)

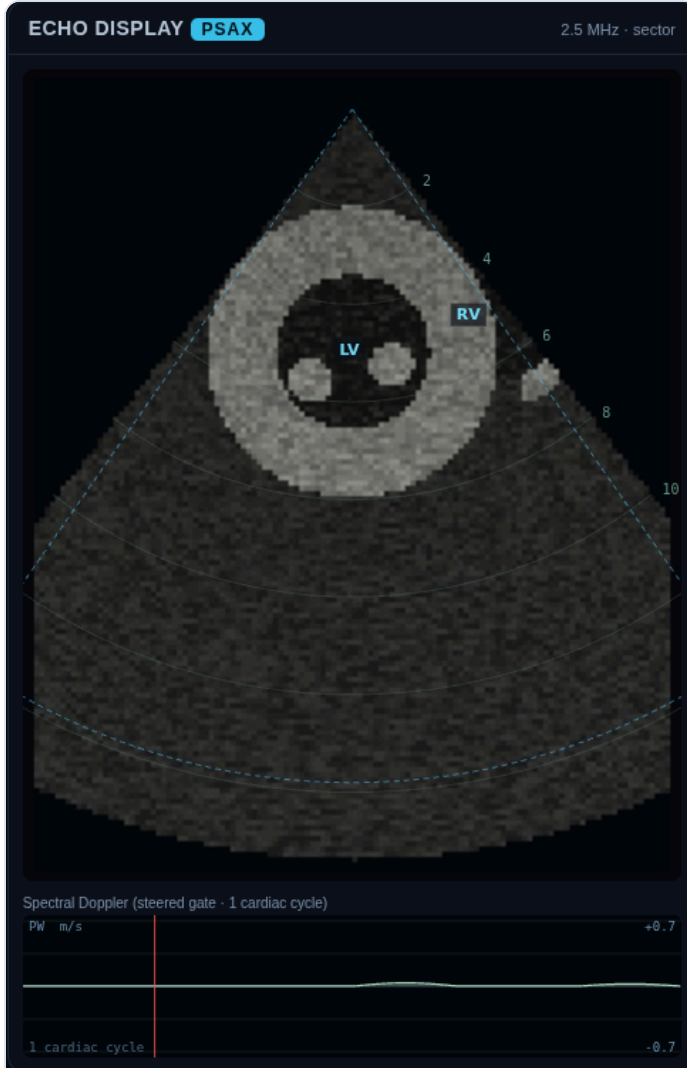
Before → after



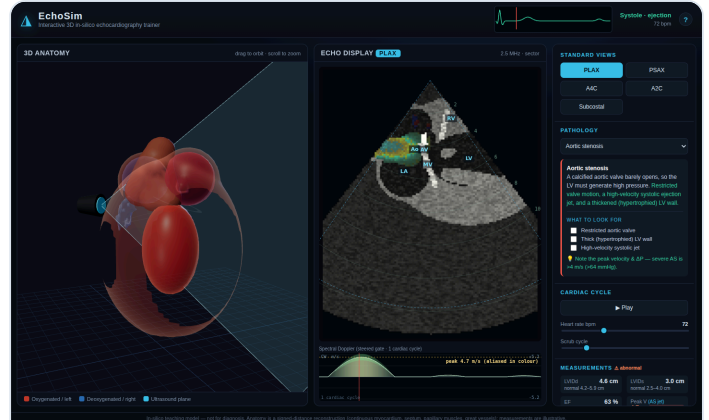
BEFORE — the original A4C: stylised ellipsoids, no continuous septum



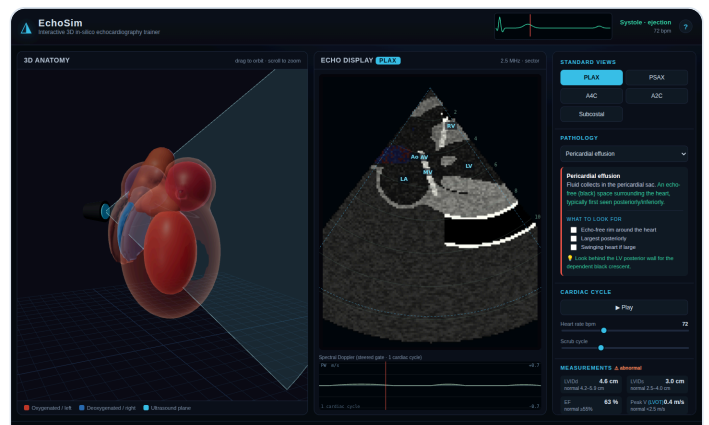
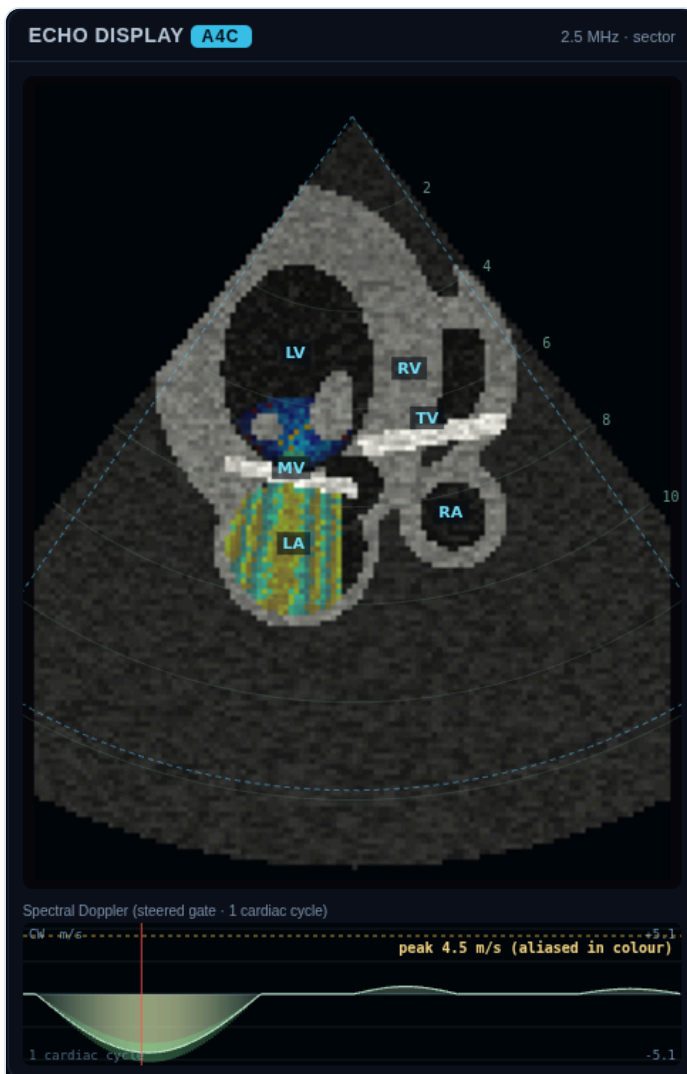
AFTER — anatomical A4C: continuous myocardium, shared septum, tricuspid apical offset, distinct-hue 3D chambers



PSAX short-axis: LV myocardial ring, both papillary muscles, septum-hugging RV crescent



Aortic stenosis PLAX: sinuses of Valsalva, LV hypertrophy, severe jet (Peak V 4.7 m/s) with BSE severity badge



Pericardial effusion: dependent echo-free crescent with a bright parietal line



Mitral regurgitation: turbulent mosaic jet confined to the LA, CW
peak 4.5 m/s (severe)

EchoSim · in-silico teaching model — not for diagnosis · calibrated to BSE/ASE chamber-quantification, normal-reference & valve-disease guidelines
(2020–2022)